

MOHAMED IBRAHIM

📍 Toronto ON | 📞 (416) 451-5682 | ✉️ mohamedhassen.ibrahim@mail.utoronto.ca
🌐 linkedin | 🐙 github | ✉️ email | 📁 portfolio

EDUCATION

University of Toronto

Honours Bachelor of Science

Toronto, Ontario

June 2028

Specialist in Computer Science (Software Engineering Stream); Minor in Statistics | GPA: 3.5+

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Machine Learning

WORK EXPERIENCE

Mozilla | Python, Rust, Django, Docker

September 2025 - August 2026

Full-Stack Software Engineer Intern

Toronto, Ontario

- Built a **versioned targeting-context pipeline** that fetched and parsed definitions from Firefox Desktop, Android, and iOS source repositories, enabling Experimenter to validate configurations against platform-specific fields and eliminating a legacy CI test that added **approximately 20 minutes** to affected pull requests.
- **Led the rebuild of Experimenter's Results page** with Django and HTMX, cutting measured load times by **more than 90%** and adding daily, weekly, and overall metric views for a platform used by **600+ internal users**.
- Fixed inconsistent enrollment-event telemetry in the **Rust Nimbus SDK**, aligning mobile and desktop reporting behavior and reducing miscategorized experiment alerts by **approximately 30%**.
- Collaborated with the Nimbus team to **decouple rollout and experiment lifecycle models**, implementing a dedicated backend state machine that eliminated overloaded state semantics and supported **45 monthly rollouts** across Firefox products serving approximately 200 million monthly active users.

Marsatek | Java, Spring Boot, PostgreSQL, React, Docker

February 2025 - April 2025

Software Developer

Remote

- Developed backend functionality for a berth-planning system using Java, Spring Boot, JPA, and PostgreSQL to manage vessel scheduling and operational data, **reducing simulated vessel wait times by 15%**.
- Created **CSV and Excel import/export workflows** with input validation, dynamic repository lookup, and reflection-based data mapping.
- Implemented **three Spring Boot REST endpoints** connecting the React frontend to PostgreSQL-backed services, with validation, persistence, and structured error handling.

PROJECTS

PaperTrail | Next.js, Supabase, Python, Gemini API, AWS App Runner, Amazon ECR

October 2025

- Built a Python receipt-ingestion pipeline supporting file and Gmail imports, OCR-based field extraction, duplicate detection, and automated folder classification, achieving **90% field-parsing accuracy** on a manually reviewed test set with an average processing time of **3 seconds per receipt**.
- Containerized the application with **Docker**, published images to **Amazon ECR**, and configured automated deployments to **AWS App Runner** through a CI/CD workflow.

Temp Tracker | React, TypeScript, Next.js, Express.js, Supabase, PostgreSQL

June 2025 - August 2025

- Created a React and Express application for importing, storing, and visualizing **10,000+ environmental temperature readings**.
- Developed a **bulk CSV ingestion pipeline** with row-level validation and immediate error reporting, allowing users to upload hundreds of temperature readings in **under 30 seconds**.
- Designed a **data-driven Supabase badge system** supporting **15+ achievement types**, allowing badge criteria to be configured without application-code changes.

TECHNICAL SKILLS

Languages: Python, Rust, Java, JavaScript, TypeScript, SQL

Frameworks: Django, Spring Boot, React, Next.js, Express.js, HTMX

Databases and Cloud: PostgreSQL, Supabase, AWS App Runner, Amazon ECR

Tools and Testing: Docker, Git, GitHub Actions, JUnit, Mockito, Jest